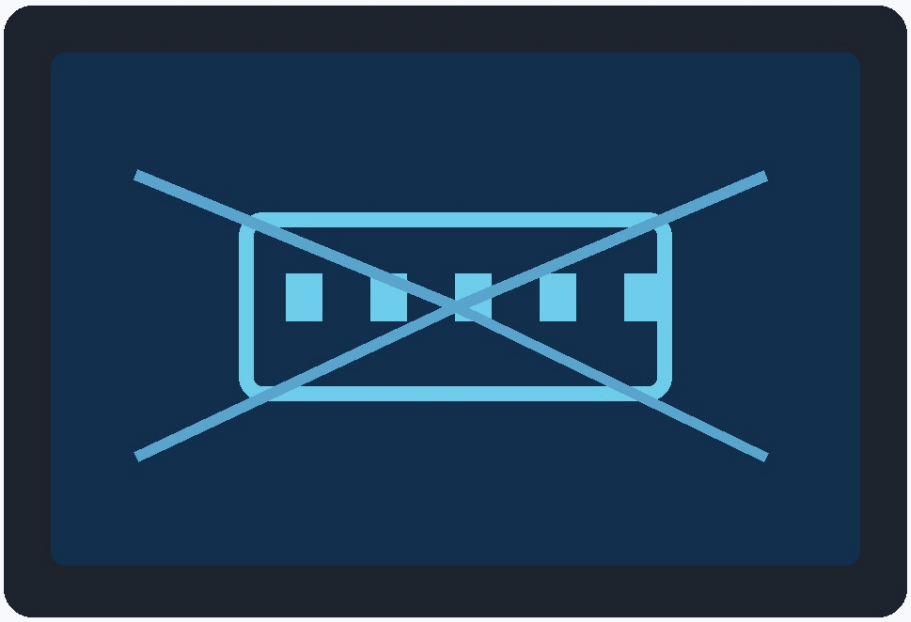
NEXASTORE

Enterprise Knowledge Library



CONNECTIVITY

NexaDock Connectivity and Docking Guide

Displays, USB, networking, power delivery and bandwidth

NexaStore Knowledge Series • 2026 Edition

1. Docking Is a Shared-Bandwidth System

A dock can combine display, USB, Ethernet, audio and charging over one host connection. The available bandwidth and power are shared, so the number of physical ports does not guarantee that every high-resolution display and high-speed peripheral can operate at maximum capability simultaneously.

Symptom	First checks
Dock not detected	Reconnect host cable; verify correct USB-C/USB4-class port; power-cycle dock
No charging	Check dock power adapter, host-port charging support and negotiated power
External display missing	Test direct display connection; confirm host display-output capability
USB device disconnects	Try another dock port; remove high-draw devices; update approved firmware
Ethernet missing	Check cable/link LEDs, OS adapter status and dock driver/firmware
Audio not available	Select dock audio device; reconnect; verify OS permissions/settings

2. Display Bandwidth

Configuration	Planning guidance
Single 1080p display	Low bandwidth pressure
Single 4K display	Requires compatible host, dock and cable
Dual 4K displays	Check dock topology and host display capability
High-refresh QHD/4K	May consume most available display bandwidth
Multiple displays + fast storage	Display and USB traffic may compete depending on dock architecture

3. Power Delivery

Observation	Meaning / action
Laptop charges slowly	Dock may provide less power than original laptop adapter
Battery drains under GPU load	System demand may exceed dock-delivered power
No charging but dock works	Host port may support data/display without charging
Dock resets under load	Check dock power supply and connected high-draw peripherals

4. Isolation Sequence

• Test the laptop and peripheral directly without the dock.

• Reconnect only the dock, then add devices one at a time.

• Use the supplied/approved host cable where possible.

• Reduce display resolution or refresh rate to test bandwidth constraints.

• Check approved dock firmware and host graphics/USB drivers.

• If only one host machine fails, investigate that host's port capability and software stack.

5. Agent Retrieval Boundaries

Question	Authoritative source
Does this laptop USB-C port support display?	Exact laptop model manual
How should I troubleshoot the dock?	This guide via RAG
What monitors are connected right now?	User/device telemetry if authorized
Is the dock under warranty?	Warranty entitlement service
Create a replacement case	Authorized support workflow

Educational notice

This fictional guide is designed for the NexaStore support-agent knowledge base.